TAY SHON DA

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PROFILE SUMMARY

Motivated undergraduate with a strong foundation in Python and TypeScript. Passionate about LLMs, Agentic AI, Computer Vision and IoT. Demonstrated ability to build practical AI applications using modern frameworks like LangChain, LangGraph, Hugging Face, and PyTorch. Seeking a full time job in Artificial Intelligence or Machine Learning for a fresh graduate to apply technical skills in contributing innovative software solutions.

EDUCATION

Bachelor of Computer Science (Software Engineering) with Honours Oct 2022 – Sep 2026
University Malaysia Pahang Al-Sultan Abdullah

- CGPA: 3.82
- Dean's List Recipient (All Semesters)

RELEVANT COURSEWORK

Machine Learning Applications | Artificial Intelligence Techniques | Applied Statistics | IoT Data Integration

SKILLS

Programming Languages
Python | TypeScript | JavaScript | Dart

Libraries & Frameworks
LangChain | LangGraph | Hugging Face | PyTorch | Scikit-Learn

Web Development Frameworks
Next.js | Laravel | Flutter | FastAPI

DevOps & Tools
Git | Linux | Docker | AWS | Node-RED

Languages
English (Fluent) | Mandarin (Fluent) | Malay (Fluent)

PROFESSIONAL EXPERIENCE

Wiwynn Corporation – *Test Engineer Intern* Mar 2026 – Aug 2026

- Maintained Python environments, Bash startup scripts, and system configuration files across distributed Linux-based Raspberry Pi edge nodes.
- Maintained dozens of Raspberry Pi edge controllers and resolved real-time network between physical test fixtures, Programmable Logic Controllers (PLCs), and automated robotic arms.
- Configured 50G speeds switches, VLAN routing, and in-band/out-of-band (BMC) network planes, and BIOS

settings.

PROJECTS

Agentic Chatbot | Personal Project

- Developed an asynchronous agentic chatbot backend using FastAPI and LangGraph with SQLite checkpointing, enabling stateful multi-turn conversations and Human-in-the-Loop (HITL) workflow interruptions.
- Engineered a thread-isolated RAG pipeline using Chroma vector database and SHA-256 deduplicated ingestion for multi-format documents (PDF, DOCX, TXT, MD, CSV) with a deterministic file-first grounding policy.
- Integrated dynamic multi-provider LLM routing across Groq and OpenRouter with real-time NDJSON event streaming, active run cancellation, and an extensible tool suite featuring Tavily web search, financial APIs, and persistent memory.

LLM Wiki Knowledge Vault & Graph | Personal Project

- Architected an agent-governed Obsidian knowledge repository integrating a compiled Markdown Wiki with Open Knowledge Format (OKF) specification under a Human-in-the-Loop approval model.
- Implemented a four-layer immutability defense (`chmod 444`, Git pre-commit hooks, path denies, pre-tool guards) to preserve raw evidence integrity and guarantee reversibility across AI agents.
- Built a graph visualization that parses Markdown Wiki files and a lightweight in-memory retrieval system using character n-gram TF-IDF vectorization and LangChain orchestration.
- Containerized the graph and retrieval system with Docker and Mangum for serverless deployment on AWS Lambda.

Deepfake Image Detection & Deployment | Final Year Project

- Performed Low-Rank Adaptation (LoRA) fine-tuning on pre-trained deepfake detection models using the Hugging Face ecosystem.
- Optimized the models for specific image datasets to improve classification accuracy.
- Containerized the application using Docker and deployed the final model to Hugging Face Spaces.